BIOLOGY ▪ ECOLOGY ▪ SECONDARY SCHOOL REVISION

NUTRIENT CYCLING

Carbon ▪ Nitrogen ▪ Water

PrepNAce
One-page revision cheatsheet

BIG IDEAS & KEY TERMS

- **Energy flows** one way (Sun → producers → consumers → heat); **matter cycles**.
- **Nutrient (biogeochemical) cycle:** path of an element through living (*bio*) and non-living (*geo*) parts of an ecosystem.
- **Reservoir (store/sink):** where a nutrient is held (air, ocean, rocks, soil). **Flux:** movement between stores.
- **Producers** make food; **consumers** eat; **decomposers** (bacteria, fungi) recycle dead matter.

MUST-KNOW EQUATIONS

Photosynthesis $6\,CO_2 + 6\,H_2O \xrightarrow{\text{light}} C_6H_{12}O_6 + 6\,O_2$

Respiration $C_6H_{12}O_6 + 6\,O_2 \longrightarrow 6\,CO_2 + 6\,H_2O + \text{energy}$

Combustion $CH_4 + 2\,O_2 \longrightarrow CO_2 + 2\,H_2O$

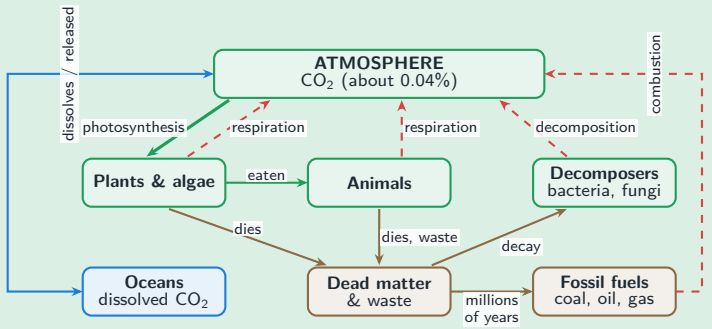
CO_2 in oceans $CO_2 + H_2O \rightleftharpoons H_2CO_3$

Haber process $N_2 + 3\,H_2 \rightleftharpoons 2\,NH_3$

Nitrification $NH_4^+ \longrightarrow NO_2^- \longrightarrow NO_3^-$

THE CARBON CYCLE

Carbon is in every organic molecule



↓ REMOVES CO_2 (green)
Photosynthesis; ocean uptake; shells and limestone ($CaCO_3$).

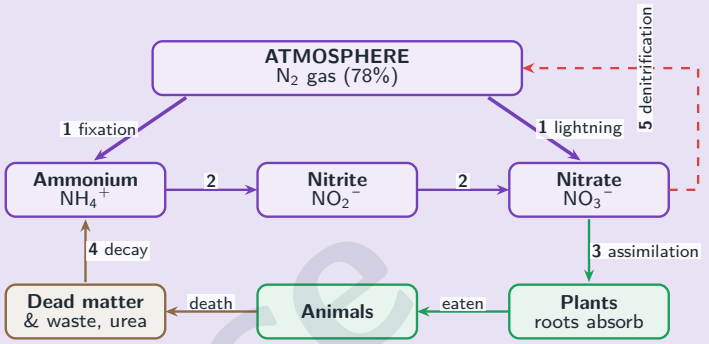
↑ ADDS CO_2 (red dashed)
Respiration; decomposition; burning fuels; volcanoes.

- **Stores:** forests, soil, oceans, limestone, fossil fuels.
- **Sink:** absorbs more CO_2 than it releases (growing forest). **Source:** releases more.

Human impact: burning fossil fuels and deforestation raise CO_2 from about 280 ppm (pre-industrial) to over 420 ppm ⇒ stronger greenhouse effect ⇒ global warming.

THE NITROGEN CYCLE

Needed for proteins, DNA, chlorophyll

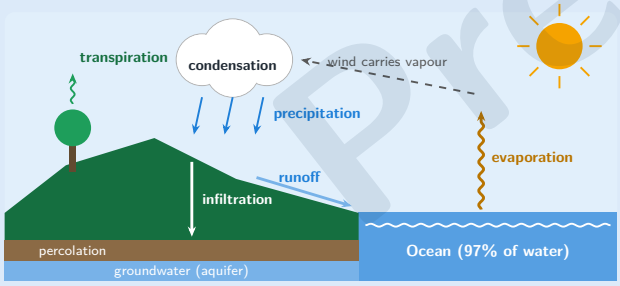


- **1 Fixation:** $N_2 \rightarrow NH_3/NO_3^-$: *Rhizobium* (legume nodules), *Azotobacter*, lightning, Haber process.
- **2 Nitrification:** $NH_4^+ \longrightarrow NO_2^- \longrightarrow NO_3^-$ (*Nitrosomonas*, *Nitrobacter*).
- **3 Assimilation:** roots absorb NO_3^- to make proteins; animals eat plants.
- **4 Ammonification:** decomposers turn dead matter and urea into NH_4^+.
- **5 Denitrification:** $NO_3^- \longrightarrow N_2$ (*Pseudomonas*; no oxygen).

Human impact: fertiliser runoff ⇒ **eutrophication** (algal bloom → low oxygen → fish die). Legumes and manure restore N.

THE WATER CYCLE

Driven by the Sun's energy and gravity



- **Evaporation:** Sun heats surface water → vapour. **Transpiration:** plants lose vapour through leaf stomata. Together: **evapotranspiration**.
- **Condensation:** vapour cools → droplets → clouds. **Precipitation:** rain, snow, sleet, hail.
- **Runoff:** flows over land into rivers and seas. **Infiltration:** soaks into soil; **percolation:** moves down to groundwater (water table).

Facts: about 97% of Earth's water is salty ocean; most fresh water is frozen in ice caps and glaciers. The cycle is closed, so total water stays constant.
Human impact: deforestation (less transpiration, more runoff and erosion), over-pumping of groundwater, pollution, melting ice.

THE THREE CYCLES AT A GLANCE

Cycle	Main reservoir	Key processes	Key organisms	Human impact
Carbon	Air (CO_2), oceans, fossil fuels, rocks	Photosynthesis, respiration, decomposition, combustion	Plants and algae take in; all organisms release	Fossil fuels and deforestation ⇒ global warming
Nitrogen	Air (N_2, 78%)	Fixation, nitrification, assimilation, ammonification, denitrification	*Rhizobium*, *Nitrosomonas*, *Nitrobacter*, *Pseudomonas*	Fertilisers ⇒ eutrophication
Water	Oceans (97%)	Evaporation, transpiration, condensation, precipitation, runoff, infiltration	Plants (transpiration)	Over-pumping, pollution, deforestation

WORKED EXAMPLE

Q: Maize yield falls each year. How does rotating with beans help?
A: Beans are legumes: *Rhizobium* in root nodules fixes N_2 into ammonium; nitrification gives NO_3^-. Maize roots absorb nitrate for proteins and chlorophyll, so less fertiliser is needed.

COMMON MISTAKES

- **Nitrification** makes nitrate; **denitrification** removes it.
- Plants respire all the time but photosynthesise only in light.
- Respiration ≠ breathing: it releases energy from glucose.
- Plants cannot fix N_2; only bacteria, lightning and industry can.

EXAM TIPS

- Name the **organism** and the **process**.
- Show direction: "from . . . to . . . ".
- **Describe** = what; **explain** = what + why.

Nitrogen steps: F-N-A-A-D
Fixation, **N**itrification, **A**ssimilation, **A**mmonification, **D**enitrification

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